

OC Core Journal Core Article v1.3.3

Ontology of Continua Core release review artifact

Alexander Yashin

Version 1.3.3

Contents

1	OC Core Journal Core Article v1.3.3	8
1.1	Publication Identity	8
1.2	Dedication	8
1.3	Acknowledgements	8
1.4	Abstract	8
1.5	Reader Contract	8
1.6	Table of Contents	8
2	Body	9
2.1	Block 2	9
2.1.1	Define Reader Contract	9
2.1.2	Define Intended Audiences	10
2.1.3	Define What OC Core Claims	11
2.1.4	Define What OC Core Does Not Claim	11
2.1.5	Define Release Scope vs Full Science Program	12
2.1.6	Define Claim Promotion, Demotion,	13
2.2	Block 3	14
2.2.1	Define The Continuum Problem	14
2.2.2	Define The Problem of Liveness	14
2.2.3	Define The Problem of Identity Through Change	15
2.2.4	Define The Problem of Boundaries	16
2.2.5	Define The Problem of Cross-Domain Formalization	16
2.2.6	Define Why Existing Frameworks Are Not Enough	17
2.2.7	Define Requirements for a Scientific Core Model	18
2.3	Block 4	18
2.3.1	Define General Systems Theory	18
2.3.2	Define Autopoiesis	19
2.3.3	Define Dynamical Systems	20
2.3.4	Define Category, Topos,	20
2.3.5	Define RAF Theory	21
2.3.6	Define Complexity, Information,	22
2.3.7	Define Causality	22
2.3.8	Define Systems Engineering	23

2.3.9	Define Comparator Method	24
2.4	Block 5	25
2.4.1	Define Claim Classes	25
2.4.2	Define Formal Proof Standard	25
2.4.3	Define Executable Semantics Standard	26
2.4.4	Define Empirical Evidence Standard	27
2.4.5	Define Simulation	27
2.4.6	Define Negative Controls	28
2.4.7	Define Source, Artifact,	29
2.4.8	Define Editorial	29
2.5	Block 7	30
2.5.1	Define The Core OC Tuple	30
2.5.2	Define Well-Formed Continua	31
2.5.3	Define Continuumness Functionals	31
2.5.4	Define Live Realizations	32
2.5.5	Define Lawful Transitions	33
2.5.6	Define Generalized Boundaries	33
2.5.7	Define Internal	34
2.5.8	Define Foundation-Level Assumptions	35
2.5.9	Define Formal Limits of the Core Model	36
2.6	Block 9	36
2.6.1	Define Theorem Inventory	36
2.6.2	Define Dependency Graph	37
2.6.3	Define Core Foundation Theorems	38
2.6.4	Define Boundary Theorems	38
2.6.5	Define Identity	39
2.6.6	Define K-Level Theorems	40
2.6.7	Define Minimality	40
2.6.8	Define Counterexample Boundaries	41
2.6.9	Define Proof Closure Summary	42
2.7	Block 10	42
2.7.1	Define Formalization Strategy	42
2.7.2	Define Lean/Lake Subset	43
2.7.3	Define Typed Structures in Lean	44
2.7.4	Define Machine-Checked Theorem Subset	44
2.7.5	Define Finite Model Semantics	45
2.7.6	Define Witness Cases	46
2.7.7	Define Tamper	47
2.7.8	Define Limits of the Formalized Subset	47
2.8	Block 11	48
2.8.1	Define Evidence Ledger	48
2.8.2	Define Mathematics Anchor	49
2.8.3	Define Physics Evidence Lane	49
2.8.4	Define Chemistry Evidence Lane	50
2.8.5	Define Biology Evidence Lane	51
2.8.6	Define Systems	51
2.8.7	Define Simulations	52
2.8.8	Define Adversarial Simulations	53

2.8.9	Define Target-Blind	53
2.8.10	Define Evidence Promotion Boundaries	54
2.9	Block 13	55
2.9.1	Define Falsifier Registry	55
2.9.2	Define Counterexample Search	55
2.9.3	Define Known Failure Modes	56
2.9.4	Define Unsupported Strong Claims	57
2.9.5	Define Demotion Rules	57
2.9.6	Define Residual Scientific Risks	58
2.9.7	Define Full-Science Background Obligations	59
2.10	Block 14	60
2.10.1	Define Novelty Criteria	60
2.10.2	Define Non-Equivalence Criteria	60
2.10.3	Define Overlap With Prior Art	61
2.10.4	Define Residual-Delta Analysis	62
2.10.5	Define Competitor-by-Competitor Comparison	62
2.10.6	Define Response to “This Is Just Another Theory”	63
2.10.7	Define Positioning of OC Core	64
2.11	Block 19	64
2.11.1	Define What target release Establishes	64
2.11.2	Define Scientific Contribution	65
2.11.3	Define What Remains Open	66
2.11.4	Define Research Roadmap	66
2.11.5	Define Relation to Future Releases	67
2.11.6	Define Practical Use by Scientists	68
2.11.7	Define Final Synthesis	68
2.12	Block 999	69
2.12.1	Core theorem route	69
2.12.2	Theorem fate	69
2.12.3	Ontological Structure of the Model	69
2.12.4	Theorem 1: Monotonicity of dimensionality	70
2.12.5	Theorem 2: Impossibility of spontaneous dimension creation	70
2.12.6	Theorem 3: Death through boundary	70
2.12.7	Theorem 9: Death as loss of cycles	70
2.12.8	Theorem 11: Threshold-expressivity incompatibility	70
2.12.9	Patch Model of Boundaries	71
2.12.10	Worked Example 1: collapse, residue,	71
2.12.11	Alternative-model competition	71
2.12.12	Theorem	71
2.12.13	Proof	72
2.12.14	Internal models	72
2.12.15	Type~IV Experiments: Internal Models	72
2.12.16	Type~III Experiments: Meta-Model Dynamics	72
2.12.17	(F10.27) Meta-model cycle collapse	72
2.12.18	(F6.14) Negative model confidence	73
2.12.19	(F6.33) Broken model cycle	73
2.12.20	(F6.42) Model-collapse at birth	73
2.12.21	(F9.28) Breakdown of the model cycle	73

2.12.22(F9.36) No emergence of formal models	73
2.12.23 Serious model families	74
2.12.24 Jets of Meta-Modelling Operators F	74
2.12.25 Jets of Categories of Models C	74
2.12.26 Patch Model Jets	74
2.12.27 Jets of Model Spaces M	74
2.12.28 Role of kk-Level Structure in the Core Model	75
2.12.29 Proto-Representation	75
2.12.30 Model Building	75
2.12.31 Theorem on Birth of a New Dimension via ()	75
2.12.32 Theorem 6 (Birth of a new dimension)	75
2.12.33 Theorem 9 (No “backward” dimensional reduction for a living continuum) . .	76
2.12.34 Theorem 4: Impossibility of evolution outside the embedding space	76
2.12.35 Theorem 5: Necessity	76
2.12.36 Theorem 6: Structural tension	76
2.12.37 Theorem 7: Universal law of complexity growth	76
2.12.38 Theorem 8: Impossibility of eternal stabilisation	77
2.12.39 Theorem 10: Monotonic growth of embedding spaces	77
2.12.40 Theorem 12: Incompleteness of embedding spaces	77
2.12.41 Proof-bearing	77
2.12.42 Theorem Roadmap	77
2.12.43 Load-bearing theorem spine	78
2.12.44 Theorem fate correction after external review	78
2.12.45 Proof Machinery, Revision Law, Competition,	78
2.12.46(F11.10) Failure of global coherence theorems	78
2.12.47 Proof obligations	78
2.12.48 Promoted theorem-native claim	79
2.12.49 Proof Construction, Verification,	79
2.12.50 Inference, Deduction,	79
2.12.51 Master Theorems of Core 1.2	79
2.12.52 Representability Theorems for Levels (K_2)K_2, (K_4)K_4, (K_8)K_8 . .	79
2.12.53 Theorem 1 (Representability of (K_2)K_2)	80
2.12.54 Theorem 2 (Representability of (K_4)K_4)	80
2.12.55 Theorem 3 (Representability of (K_8)K_8)	80
2.12.56 Compatibility Theorem (K M)K M	80
2.12.57 Theorem 4 (Necessary	80
2.12.58 Theorem on Impossibility of Evolution Outside (M)M	81
2.12.59 Theorem 5 (Impossibility of evolution outside the meta-domain)	81
2.12.60 Theorem on Death of a Continuum	81
2.12.61 Theorem 7 (Equivalence of death conditions)	81
2.12.62 Theorem on Phase Transition	81
2.12.63 Theorem on Impossibility of Degradational Evolution	82
2.12.64 Source Witnessing, Theorem Fate,	82
2.12.65 Executable Finite-Model Evidence	82
2.12.66 Prediction	82
2.12.67 Hybrid-Escalation Evidence Bar	82
2.12.68 Global evidence bar	83
2.12.69(P10.3) Proof-Flow Emergence	83

2.12.70	P4: Predictions for Proof/Computation Flows J^{10}	83
2.12.71	(P10.12) Proof Flows Have Attractor Structure	83
2.12.72	P1: Predictions About the Birth of Model-Spaces	83
2.12.73	(P11.2) Model-Space Dimensional Trigger	84
2.12.74	(P11.5) Universal Axes of Model-Spaces	84
2.12.75	(P11.6) Existence of Model-Topologies	84
2.12.76	(P11.17) The Model-Theoretic Cycle	84
2.12.77	(P11.21) Model-Space Fragmentation Reduces $k(K_{11})$	84
2.12.78	(P12.1) Dynamization Threshold of Model-Spaces	85
2.12.79	(P12.24) Fragmentation of Model-Dynamics Reduces $k(K_{12})$	85
2.12.80	(P12.26) No Stable Termination of Model-Dynamics	85
2.12.81	(P9.12) Proof-like Flows Are Emergent	85
2.12.82	(P9.26) Birth of Proof-Flow Attractors	85
2.12.83	Target-Blind Numeric Evidence	86
2.12.84	OC 1.3.3 Comparator	86
2.12.85	Prior-Art Interpretation for Article Readers	86
2.12.86	Prior-Art Comparator	86
2.12.87	Prior-Art	86
2.12.88	Master Monograph Appendix: Comparator And Novelty Matrix	87
2.12.89	OC133-NOVELTY-001	87
2.12.90	appendix/OC_1_3_3_COMPARATOR_AND_NOVELTY_MATRIX.tex	87
2.12.91	Reviewer challenge 7: Prior Art Positioning	87
2.12.92	Prior-Art Historian	87
2.12.93	Worked Attack Transcript C: Prior-Art Surface	88
2.12.94	Reviewer challenge 6: Phenomenon Coverage Model Card	88
2.12.95	Practical Utility	88
2.12.96	Serious model comparison matrix	88
2.12.97	Serious model comparison trace matrix	88
2.12.98	10. T133-KLEVEL: Declared adjacent K-level atlas/evaluator consistency theorem	89
2.12.99	Reviewer challenge 1: Theorem Proof Binding	89
2.12.100	Reviewer challenge 2: Theorem Public Surface Binding	89
2.12.101	Attack Family 5: Public Wording Outruns Evidence	89
2.12.102	Evidence-to-Wording Calibration	89
2.12.103	Claim-Family Evidence Recipes	90
2.12.104	Worked Attack Transcript A: Theorem Surface	90
2.12.105	Proof Machinery Appendix	90
2.12.106	Alternative-model competition matrix	90
2.12.107	Load-bearing proof tasks	90
2.12.108	Closure priority board	91
2.12.109	Ontological Structure of the Model	91
2.12.110	Theorem (global verdict-invariant minimality)	91
2.12.111	Proof	91
2.12.112	Theorem 1: Monotonicity of dimensionality	91
2.12.113	Theorem 2: Impossibility of spontaneous dimension creation	92
2.12.114	Theorem 3: Death through boundary	92
2.12.115	Theorem 4: Impossibility of evolution outside the embedding space	92
2.12.116	Theorem 5: Necessity	92

2.12.11	Theorem 6: Structural tension	92
2.12.11	Theorem 7: Universal law of complexity growth	93
2.12.11	Theorem 8: Impossibility of eternal stabilisation	93
2.12.12	Theorem 9: Death as loss of cycles	93
2.12.12	Theorem 10: Monotonic growth of embedding spaces	93
2.12.12	Theorem 11: Threshold-expressivity incompatibility	93
2.12.12	Theorem 12: Incompleteness of embedding spaces	94
2.12.12	Patch Model of Boundaries	94
2.12.12	Proof-bearing	94
2.12.12	Theorem Roadmap	94
2.12.12	Load-bearing theorem spine	94
2.12.12	Theorem fate correction after external review	95
2.12.12	Worked Example 1: collapse, residue,	95
2.12.13	Proof Machinery, Revision Law, Competition,	95
2.12.13	Alternative-model competition	95
2.12.13	Internal models	95
2.12.13	Type-IV Experiments: Internal Models	96
2.12.13	Type-III Experiments: Meta-Model Dynamics	96
2.12.13	(F10.27) Meta-model cycle collapse	96
2.12.13	(F11.10) Failure of global coherence theorems	96
2.12.13	(F6.14) Negative model confidence	96
2.12.13	(F6.33) Broken model cycle	96
2.12.13	(F6.42) Model-collapse at birth	97
2.12.14	(F9.15) Inability to integrate cross-domain evidence	97
2.12.14	(F9.28) Breakdown of the model cycle	97
2.12.14	(F9.36) No emergence of formal models	97
2.12.14	Serious model families	97
2.12.14	What prior science could do, what remained fragmented,	98
2.12.14	Proof obligations	98
2.12.14	Promoted theorem-native claim	98
2.12.14	Jets of Meta-Modelling Operators F	98
2.12.14	Jets of Categories of Models C	98
2.12.14	Patch Model Jets	99
2.12.15	Jets of Model Spaces M	99
2.12.15	Role of kk -Level Structure in the Core Model	99
2.12.15	Proof Construction, Verification,	99
2.12.15	Proto-Representation	99
2.12.15	Model Building	99
2.12.15	Inference, Deduction,	100
2.12.15	Master Theorems of Core 1.2	100
2.12.15	Representability Theorems for Levels $(K_2)K_2$, $(K_4)K_4$, $(K_8)K_8$. .	100
2.12.15	Theorem 1 (Representability of $(K_2)K_2$)	100
2.12.15	Theorem 2 (Representability of $(K_4)K_4$)	100
2.12.16	Theorem 3 (Representability of $(K_8)K_8$)	101
2.12.16	Compatibility Theorem $(K\ M)K\ M$	101
2.12.16	Theorem 4 (Necessary	101
2.12.16	Theorem on Impossibility of Evolution Outside $(M)M$	101
2.12.16	Theorem 5 (Impossibility of evolution outside the meta-domain)	102

2.12.165	Theorem on Birth of a New Dimension via ()	102
2.12.166	Theorem 6 (Birth of a new dimension)	102
2.12.167	Theorem on Death of a Continuum	102
2.12.168	Theorem 7 (Equivalence of death conditions)	102
2.12.169	Theorem on Phase Transition	103
2.12.170	Theorem on Impossibility of Degradational Evolution	103
2.12.171	Theorem 9 (No “backward” dimensional reduction for a living continuum)	103
2.12.172	Evidence Summary	103
2.12.173	Theorem	103
2.12.174.	T133-K0-RES: K0 resolution-relative distinguishability theorem	104
2.12.175.	T133-OMEGA-STATUS: Typed liveness, death, residue,	104
2.12.176.	T133-K-ZERO: Continuumness zero obstruction theorem	104
2.12.177.	T133-BOUNDARY: Metric-threshold boundary specialization theorem	104
2.12.178.	T133-HYBRID: Typed update	104
2.12.179.	T133-DIM: Historical axis	105
2.12.180.	T133-CYCLE: Live-status cycle-mode requirement theorem	105
2.12.181.	T133-ID: Identity, residue,	105
2.12.182.	T133-MIN: Declared semantic-verdict component independence theorem	105
2.12.183	Does OC actually explain operators in non-smooth proof	106
2.12.184	Does OC actually explain theory change as live claim/evidence update?	106
2.12.185	Theorem	106
2.12.186	Master Monograph Appendix: Boundary Representation Theorem	106
2.12.187	Boundary Representation Theorem	106
2.12.188	Master Monograph Appendix: Verdict Invariant Minimality Full Proof	107
2.12.189	OC 1.3.3 Verdict-Invariant Minimality Full Proof	107
2.12.190	FM-T133-HYBRID-PROOF-UPDATE-POS	107
2.12.191	FM-T133-HYBRID-PROOF-UPDATE-NEG	107
2.12.192	FM-T133-HYBRID-PROOF-NO-RULE-NEG	107
2.12.193	appendix/OC_1_3_3_BOUNDARY_REPRESENTATION_THEOREM.tex	108
2.12.194	appendix/OC_1_3_3_VERDICT_INVARIANT_MINIMALITY_FULL_PROOF.tex	108
2.12.195	Evidence Inputs	108
2.12.196	Evidence Reading Examples	108
2.12.197	Proof Machinery Appendix	108
2.12.198	Alternative-model competition matrix	109
2.12.199	Boundary Representation Theorem	109
2.12.200	Theorem	109
2.12.201	Proof	109
2.12.202	OC 1.3.3 Verdict-Invariant Minimality Full Proof	109
2.12.203	Load-bearing proof obligations	110
2.12.204	Closure priority board	110
2.12.205	Definitions Before Evidence IDs	110
2.12.206	Evidence	110
2.12.207	Results	110
2.12.208	(F9.15) Inability to integrate cross-domain evidence	111
2.12.209	What prior science could do, what remained fragmented,	111
2.13	Source Trace	111

1 OC Core Journal Core Article v1.3.3

1.1 Publication Identity

Author. Alexander Yashin, Independent Researcher, ORCID 0009-0008-6166-0914. **Research instrument.** Logion is the research-instrument and institute-automation system used to prepare, check, package, and audit the work; it is not an author. **Methodological framework.** ESTRA is the methodological framework used in the work; it is not an author or affiliation. **Version.** 1.3.3. **Release identity.** oc_core_1_3_3. **Concept DOI.** 10.5281/zenodo.17899134. **Artifact role.** Compact article-style scientific argument. **Intended audience.** journal editors, reviewers, and scientists who need a compact entry point before the full monograph.

1.2 Dedication

Dedicated to my dear wife Maria, without whom this work would have been impossible.

1.3 Acknowledgements

Substantive review and idea acknowledgements. They are acknowledged for review pressure, ideas, criticism, or external response that improved the work. Acknowledgement does not imply authorship, endorsement, publication approval, or agreement with the theory's release form.

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1.4 Abstract

This article gives the compact review route through the model core. It states the scientific problem, the formal contribution, the evidence and falsification route, novelty boundaries, and the claims that are promoted for review.

1.5 Reader Contract

Read this document as a compressed argument. It is designed to be checked against the monograph and companion artifacts, not as an isolated replacement for them. The release promotes evidence-bound model-core claims and excludes unsupported complete-science closure, unrestricted all-domain numerical completion, or unrestricted superiority-over-modern-science claims.

1.6 Table of Contents

The generated PDF contents provide page locations. The reader-facing route is:

1. Publication identity
2. Dedication
3. Acknowledgements
4. Abstract
5. Reader contract
6. Reading map

7. Block 2: Define Reader Contract
8. Block 2: Define Intended Audiences
9. Block 2: Define What OC Core Claims
10. Block 2: Define What OC Core Does Not Claim
11. Block 2: Define Release Scope vs Full Science Program
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37. Block 7: Define The Core OC Tuple
38. Block 7: Define Well-Formed Continua
39. Block 7: Define Continuumness Functionals
40. Block 7: Define Live Realizations
41. Additional sections continue in document order.

2 Body

2.1 Block 2

2.1.1 Define Reader Contract

Define Reader Contract is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Reader Contract. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support

route instead of relying on assertion.

Bind Reader Contract to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Reader Contract states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Reader Contract synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.1.2 Define Intended Audiences

Define Intended Audiences and Reading Paths is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Intended Audiences and Reading Paths. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Intended Audiences and Reading Paths synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.1.3 Define What OC Core Claims

Define What OC Core Claims is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for What OC Core Claims. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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2.1.4 Define What OC Core Does Not Claim

Define What OC Core Does Not Claim is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for What OC Core Does Not Claim. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed

to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind What OC Core Does Not Claim to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for What OC Core Does Not Claim states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize What OC Core Does Not Claim synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.1.5 Define Release Scope vs Full Science Program

Define Release Scope vs Full Science Program is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Release Scope vs Full Science Program. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Release Scope vs Full Science Program to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Release Scope vs Full Science Program states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer

objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Release Scope vs Full Science Program synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.1.6 Define Claim Promotion, Demotion,

Define Claim Promotion, Demotion, and Background Obligations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Claim Promotion, Demotion, and Background Obligations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Claim Promotion, Demotion, and Background Obligations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Claim Promotion, Demotion, and Background Obligations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Claim Promotion, Demotion, and Background Obligations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2 Block 3

2.2.1 Define The Continuum Problem

Define The Continuum Problem is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for The Continuum Problem. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind The Continuum Problem to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for The Continuum Problem states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize The Continuum Problem synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.2 Define The Problem of Liveness

Define The Problem of Liveness and Death is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for The Problem of Liveness and Death. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind The Problem of Liveness and Death to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The

support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for The Problem of Liveness and Death states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize The Problem of Liveness and Death synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.3 Define The Problem of Identity Through Change

Define The Problem of Identity Through Change is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for The Problem of Identity Through Change. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind The Problem of Identity Through Change to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for The Problem of Identity Through Change states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize The Problem of Identity Through Change synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains:

may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.4 Define The Problem of Boundaries

Define The Problem of Boundaries is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for The Problem of Boundaries. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind The Problem of Boundaries to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for The Problem of Boundaries states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize The Problem of Boundaries synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.5 Define The Problem of Cross-Domain Formalization

Define The Problem of Cross-Domain Formalization is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for The Problem of Cross-Domain Formalization. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind The Problem of Cross-Domain Formalization to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route

uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for The Problem of Cross-Domain Formalization states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize The Problem of Cross-Domain Formalization synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.6 Define Why Existing Frameworks Are Not Enough

Define Why Existing Frameworks Are Not Enough is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Why Existing Frameworks Are Not Enough. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Why Existing Frameworks Are Not Enough to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Why Existing Frameworks Are Not Enough states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Why Existing Frameworks Are Not Enough synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The

synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.7 Define Requirements for a Scientific Core Model

Define Requirements for a Scientific Core Model is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Requirements for a Scientific Core Model. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Requirements for a Scientific Core Model to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Requirements for a Scientific Core Model states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Requirements for a Scientific Core Model synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3 Block 4

2.3.1 Define General Systems Theory

Define General Systems Theory is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for General Systems Theory. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind General Systems Theory to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for General Systems Theory states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize General Systems Theory synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.2 Define Autopoiesis

Define Autopoiesis and Organizational Closure is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Autopoiesis and Organizational Closure. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Autopoiesis and Organizational Closure to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Autopoiesis and Organizational Closure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Autopoiesis and Organizational Closure synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.3 Define Dynamical Systems

Define Dynamical Systems and Control Theory is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Dynamical Systems and Control Theory. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Dynamical Systems and Control Theory to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Dynamical Systems and Control Theory states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Dynamical Systems and Control Theory synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.4 Define Category, Topos,

Define Category, Topos, and Formal-Ontology Approaches is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Category, Topos, and Formal-Ontology Approaches. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Category, Topos, and Formal-Ontology Approaches to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Category, Topos, and Formal-Ontology Approaches states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Category, Topos, and Formal-Ontology Approaches synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.5 Define RAF Theory

Define RAF Theory and Origin-of-Life Formalisms is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for RAF Theory and Origin-of-Life Formalisms. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind RAF Theory and Origin-of-Life Formalisms to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for RAF Theory and Origin-of-Life Formalisms states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may

synthesize the local consequence without inflating it.

Synthesize RAF Theory and Origin-of-Life Formalisms synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.6 Define Complexity, Information,

Define Complexity, Information, and Emergence Measures is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Complexity, Information, and Emergence Measures. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Complexity, Information, and Emergence Measures to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Complexity, Information, and Emergence Measures states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Complexity, Information, and Emergence Measures synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.7 Define Causality

Define Causality and Identity Theories is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Causality and Identity Theories. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed

to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Causality and Identity Theories to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Causality and Identity Theories states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Causality and Identity Theories synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.8 Define Systems Engineering

Define Systems Engineering and Institutional Models is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Systems Engineering and Institutional Models. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Systems Engineering and Institutional Models to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Systems Engineering and Institutional Models states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer

objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Systems Engineering and Institutional Models synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.9 Define Comparator Method

Define Comparator Method and Positioning Map is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Comparator Method and Positioning Map. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Comparator Method and Positioning Map to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Comparator Method and Positioning Map states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Comparator Method and Positioning Map synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4 Block 5

2.4.1 Define Claim Classes

Define Claim Classes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Claim Classes. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Claim Classes to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Claim Classes states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Claim Classes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.2 Define Formal Proof Standard

Define Formal Proof Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Formal Proof Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Formal Proof Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is

promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Formal Proof Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Formal Proof Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.3 Define Executable Semantics Standard

Define Executable Semantics Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Executable Semantics Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Executable Semantics Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Executable Semantics Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Executable Semantics Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize

established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.4 Define Empirical Evidence Standard

Define Empirical Evidence Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Empirical Evidence Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Empirical Evidence Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Empirical Evidence Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Empirical Evidence Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.5 Define Simulation

Define Simulation and Counterexample Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Simulation and Counterexample Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Simulation and Counterexample Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route

uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Simulation and Counterexample Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Simulation and Counterexample Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.6 Define Negative Controls

Define Negative Controls and Falsifiers is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Negative Controls and Falsifiers. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Negative Controls and Falsifiers to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Negative Controls and Falsifiers states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Negative Controls and Falsifiers synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws

on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.7 Define Source, Artifact,

Define Source, Artifact, and Traceability Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Source, Artifact, and Traceability Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Source, Artifact, and Traceability Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Source, Artifact, and Traceability Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Source, Artifact, and Traceability Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.8 Define Editorial

Define Editorial and Reviewer Evidence Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Editorial and Reviewer Evidence Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Editorial and Reviewer Evidence Standard to proof, evidence, or replay carries the evidential

burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Editorial and Reviewer Evidence Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Editorial and Reviewer Evidence Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5 Block 7

2.5.1 Define The Core OC Tuple

Define The Core OC Tuple is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for The Core OC Tuple. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind The Core OC Tuple to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for The Core OC Tuple states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local

consequence without inflating it.

Synthesize The Core OC Tuple synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.2 Define Well-Formed Continua

Define Well-Formed Continua is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Well-Formed Continua. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Well-Formed Continua to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Well-Formed Continua states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Well-Formed Continua synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.3 Define Continuumness Functionals

Define Continuumness Functionals is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Continuumness Functionals. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or

universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Continuumness Functionals to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Continuumness Functionals states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Continuumness Functionals synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.4 Define Live Realizations

Define Live Realizations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Live Realizations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Live Realizations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Live Realizations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing bound-

ary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Live Realizations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.5 Define Lawful Transitions

Define Lawful Transitions is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Lawful Transitions. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Lawful Transitions to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Lawful Transitions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Lawful Transitions synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.6 Define Generalized Boundaries

Define Generalized Boundaries is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Generalized Boundaries. The paragraph draws on formal model and foundation and states

the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Generalized Boundaries to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Generalized Boundaries states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Generalized Boundaries synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.7 Define Internal

Define Internal and External Relations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Internal and External Relations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Internal and External Relations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Internal and External Relations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence

as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Internal and External Relations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.8 Define Foundation-Level Assumptions

Define Foundation-Level Assumptions is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Foundation-Level Assumptions. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Foundation-Level Assumptions to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Foundation-Level Assumptions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Foundation-Level Assumptions synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.9 Define Formal Limits of the Core Model

Define Formal Limits of the Core Model is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Formal Limits of the Core Model. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Formal Limits of the Core Model to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Formal Limits of the Core Model states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Formal Limits of the Core Model synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6 Block 9

2.6.1 Define Theorem Inventory

Define Theorem Inventory is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Theorem Inventory. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Theorem Inventory to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is

promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Theorem Inventory states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Theorem Inventory synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6.2 Define Dependency Graph

Define Dependency Graph is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Dependency Graph. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Dependency Graph to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Dependency Graph states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Dependency Graph synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize

established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6.3 Define Core Foundation Theorems

Define Core Foundation Theorems is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Core Foundation Theorems. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Core Foundation Theorems to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Core Foundation Theorems states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Core Foundation Theorems synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6.4 Define Boundary Theorems

Define Boundary Theorems is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Boundary Theorems. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Boundary Theorems to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empiri-

cal computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Boundary Theorems states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Boundary Theorems synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6.5 Define Identity

Define Identity and Rebirth Theorems is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Identity and Rebirth Theorems. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Identity and Rebirth Theorems to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Identity and Rebirth Theorems states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Identity and Rebirth Theorems synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws

on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6.6 Define K-Level Theorems

Define K-Level Theorems is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for K-Level Theorems. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind K-Level Theorems to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for K-Level Theorems states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize K-Level Theorems synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6.7 Define Minimality

Define Minimality and Independence Results is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Minimality and Independence Results. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Minimality and Independence Results to proof, evidence, or replay carries the evidential

burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Minimality and Independence Results states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Minimality and Independence Results synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6.8 Define Counterexample Boundaries

Define Counterexample Boundaries is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Counterexample Boundaries. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Counterexample Boundaries to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Counterexample Boundaries states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Counterexample Boundaries synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6.9 Define Proof Closure Summary

Define Proof Closure Summary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Proof Closure Summary. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Proof Closure Summary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Proof Closure Summary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Proof Closure Summary synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7 Block 10

2.7.1 Define Formalization Strategy

Define Formalization Strategy is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Formalization Strategy. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims

without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Formalization Strategy to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Formalization Strategy states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Formalization Strategy synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7.2 Define Lean/Lake Subset

Define Lean/Lake Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Lean/Lake Subset. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Lean/Lake Subset to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Lean/Lake Subset states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The gov-

erning boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Lean/Lake Subset synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7.3 Define Typed Structures in Lean

Define Typed Structures in Lean is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Typed Structures in Lean. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Typed Structures in Lean to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Typed Structures in Lean states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Typed Structures in Lean synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7.4 Define Machine-Checked Theorem Subset

Define Machine-Checked Theorem Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Machine-Checked Theorem Subset. The paragraph draws on formal

model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Machine-Checked Theorem Subset to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Machine-Checked Theorem Subset states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Machine-Checked Theorem Subset synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7.5 Define Finite Model Semantics

Define Finite Model Semantics is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Finite Model Semantics. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Finite Model Semantics to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Finite Model Semantics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as

permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Finite Model Semantics synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7.6 Define Witness Cases

Define Witness Cases is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Witness Cases. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Witness Cases to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Witness Cases states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Witness Cases synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7.7 Define Tamper

Define Tamper and Negative Controls is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Tamper and Negative Controls. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Tamper and Negative Controls to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Tamper and Negative Controls states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Tamper and Negative Controls synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7.8 Define Limits of the Formalized Subset

Define Limits of the Formalized Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Limits of the Formalized Subset. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Limits of the Formalized Subset to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited

proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Limits of the Formalized Subset states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Limits of the Formalized Subset synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8 Block 11

2.8.1 Define Evidence Ledger

Define Evidence Ledger is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Evidence Ledger. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Evidence Ledger to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Evidence Ledger states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Evidence Ledger synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize

established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.2 Define Mathematics Anchor

Define Mathematics Anchor is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Mathematics Anchor. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Mathematics Anchor to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Mathematics Anchor states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Mathematics Anchor synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.3 Define Physics Evidence Lane

Define Physics Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Physics Evidence Lane. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Physics Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical

computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Physics Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Physics Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.4 Define Chemistry Evidence Lane

Define Chemistry Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Chemistry Evidence Lane. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Chemistry Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Chemistry Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Chemistry Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on

reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.5 Define Biology Evidence Lane

Define Biology Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Biology Evidence Lane. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Biology Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Biology Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Biology Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.6 Define Systems

Define Systems and Civilizational Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Systems and Civilizational Evidence Lane. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Systems and Civilizational Evidence Lane to proof, evidence, or replay carries the evidential

burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Systems and Civilizational Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Systems and Civilizational Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.7 Define Simulations

Define Simulations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Simulations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Simulations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Simulations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Simulations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.8 Define Adversarial Simulations

Define Adversarial Simulations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Adversarial Simulations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Adversarial Simulations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Adversarial Simulations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Adversarial Simulations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.9 Define Target-Blind

Define Target-Blind and Held-Out Evidence is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Target-Blind and Held-Out Evidence. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Target-Blind and Held-Out Evidence to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Target-Blind and Held-Out Evidence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Target-Blind and Held-Out Evidence synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.8.10 Define Evidence Promotion Boundaries

Define Evidence Promotion Boundaries is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Evidence Promotion Boundaries. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Evidence Promotion Boundaries to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Evidence Promotion Boundaries states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Evidence Promotion Boundaries synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.9 Block 13

2.9.1 Define Falsifier Registry

Define Falsifier Registry is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Falsifier Registry. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Falsifier Registry to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Falsifier Registry states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Falsifier Registry synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.9.2 Define Counterexample Search

Define Counterexample Search is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Counterexample Search. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims

without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Counterexample Search to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Counterexample Search states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Counterexample Search synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.9.3 Define Known Failure Modes

Define Known Failure Modes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Known Failure Modes. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Known Failure Modes to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Known Failure Modes states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The gov-

erning boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Known Failure Modes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.9.4 Define Unsupported Strong Claims

Define Unsupported Strong Claims is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Unsupported Strong Claims. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Unsupported Strong Claims to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Unsupported Strong Claims states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Unsupported Strong Claims synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.9.5 Define Demotion Rules

Define Demotion Rules is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Demotion Rules. The paragraph draws on formal model and foundation and states the local

vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Demotion Rules to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Demotion Rules states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Demotion Rules synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.9.6 Define Residual Scientific Risks

Define Residual Scientific Risks is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Residual Scientific Risks. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Residual Scientific Risks to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Residual Scientific Risks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as

permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Residual Scientific Risks synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.9.7 Define Full-Science Background Obligations

Define Full-Science Background Obligations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Full-Science Background Obligations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Full-Science Background Obligations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Full-Science Background Obligations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Full-Science Background Obligations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.10 Block 14

2.10.1 Define Novelty Criteria

Define Novelty Criteria is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Novelty Criteria. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Novelty Criteria to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Novelty Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Novelty Criteria synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.10.2 Define Non-Equivalence Criteria

Define Non-Equivalence Criteria is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Non-Equivalence Criteria. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Non-Equivalence Criteria to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is

promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Non-Equivalence Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Non-Equivalence Criteria synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.10.3 Define Overlap With Prior Art

Define Overlap With Prior Art is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Overlap With Prior Art. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Overlap With Prior Art to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Overlap With Prior Art states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Overlap With Prior Art synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize

established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.10.4 Define Residual-Delta Analysis

Define Residual-Delta Analysis is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Residual-Delta Analysis. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Residual-Delta Analysis to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Residual-Delta Analysis states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Residual-Delta Analysis synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.10.5 Define Competitor-by-Competitor Comparison

Define Competitor-by-Competitor Comparison is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Competitor-by-Competitor Comparison. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Competitor-by-Competitor Comparison to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route

uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Competitor-by-Competitor Comparison states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Competitor-by-Competitor Comparison synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.10.6 Define Response to “This Is Just Another Theory”

Define Response to “This Is Just Another Theory” is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Response to “This Is Just Another Theory”. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Response to “This Is Just Another Theory” to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Response to “This Is Just Another Theory” states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Response to “This Is Just Another Theory” synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The

synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.10.7 Define Positioning of OC Core

Define Positioning of OC Core is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Positioning of OC Core. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Positioning of OC Core to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Positioning of OC Core states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Positioning of OC Core synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.11 Block 19

2.11.1 Define What target release Establishes

Define What target release Establishes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for What target release Establishes. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind What target release Establishes to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for What target release Establishes states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize What target release Establishes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.11.2 Define Scientific Contribution

Define Scientific Contribution is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Scientific Contribution. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Scientific Contribution to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Scientific Contribution states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Scientific Contribution synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.11.3 Define What Remains Open

Define What Remains Open is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for What Remains Open. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind What Remains Open to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for What Remains Open states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize What Remains Open synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.11.4 Define Research Roadmap

Define Research Roadmap is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Research Roadmap. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Research Roadmap to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Research Roadmap states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Research Roadmap synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.11.5 Define Relation to Future Releases

Define Relation to Future Releases is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Relation to Future Releases. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Relation to Future Releases to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Relation to Future Releases states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Relation to Future Releases synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.11.6 Define Practical Use by Scientists

Define Practical Use by Scientists and Institutions is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Practical Use by Scientists and Institutions. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Practical Use by Scientists and Institutions to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Practical Use by Scientists and Institutions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Practical Use by Scientists and Institutions synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.11.7 Define Final Synthesis

Define Final Synthesis is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Final Synthesis. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Final Synthesis to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Final Synthesis states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Final Synthesis synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.12 Block 999

2.12.1 Core theorem route

Core theorem route and scope carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.2 Theorem fate

Theorem fate and scope discipline carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.3 Ontological Structure of the Model

Ontological Structure of the Model is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for

Ontological Structure of the Model. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.4 Theorem 1: Monotonicity of dimensionality

Theorem 1: Monotonicity of dimensionality carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.5 Theorem 2: Impossibility of spontaneous dimension creation

Theorem 2: Impossibility of spontaneous dimension creation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.6 Theorem 3: Death through boundary

Theorem 3: Death through boundary and embedding collapse carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.7 Theorem 9: Death as loss of cycles

Theorem 9: Death as loss of cycles carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.8 Theorem 11: Threshold-expressivity incompatibility

Theorem 11: Threshold-expressivity incompatibility carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics,

replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.9 Patch Model of Boundaries

Patch Model of Boundaries is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Patch Model of Boundaries. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.10 Worked Example 1: collapse, residue,

Worked Example 1: collapse, residue, and rebirth in a minimal admissible-state model is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Worked Example 1: collapse, residue, and rebirth in a minimal admissible-state model. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.11 Alternative-model competition

Alternative-model competition is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Alternative-model competition. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.12 Theorem

Theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.13 Proof

Proof carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.14 Internal models

Internal models is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Internal models. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.15 Type~IV Experiments: Internal Models

Type~IV Experiments: Internal Models and Coherence Testing is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Type~IV Experiments: Internal Models and Coherence Testing. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.16 Type~III Experiments: Meta-Model Dynamics

Type~III Experiments: Meta-Model Dynamics and Explanatory Power is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Type~III Experiments: Meta-Model Dynamics and Explanatory Power. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.17 (F10.27) Meta-model cycle collapse

(F10.27) Meta-model cycle collapse is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F10.27) Meta-model cycle collapse. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.18 (F6.14) Negative model confidence

(F6.14) Negative model confidence is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F6.14) Negative model confidence. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.19 (F6.33) Broken model cycle

(F6.33) Broken model cycle is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F6.33) Broken model cycle. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.20 (F6.42) Model-collapse at birth

(F6.42) Model-collapse at birth is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F6.42) Model-collapse at birth. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.21 (F9.28) Breakdown of the model cycle

(F9.28) Breakdown of the model cycle is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F9.28) Breakdown of the model cycle. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.22 (F9.36) No emergence of formal models

(F9.36) No emergence of formal models is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F9.36) No emergence of formal models. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.23 Serious model families

Serious model families is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Serious model families. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.24 Jets of Meta-Modelling Operators F

Jets of Meta-Modelling Operators F is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Jets of Meta-Modelling Operators F. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.25 Jets of Categories of Models C

Jets of Categories of Models C is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Jets of Categories of Models C. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.26 Patch Model Jets

Patch Model Jets and Local Instabilities is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Patch Model Jets and Local Instabilities. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.27 Jets of Model Spaces M

Jets of Model Spaces M is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Jets of Model Spaces M. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.28 Role of kk-Level Structure in the Core Model

Role of kk-Level Structure in the Core Model is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Role of kk-Level Structure in the Core Model. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.29 Proto-Representation

Proto-Representation and Internal Modelling Processes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Proto-Representation and Internal Modelling Processes. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.30 Model Building

Model Building and Theorisation Processes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Model Building and Theorisation Processes. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.31 Theorem on Birth of a New Dimension via ()

Theorem on Birth of a New Dimension via () carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.32 Theorem 6 (Birth of a new dimension)

Theorem 6 (Birth of a new dimension) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.33 Theorem 9 (No “backward” dimensional reduction for a living continuum)

Theorem 9 (No “backward” dimensional reduction for a living continuum) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.34 Theorem 4: Impossibility of evolution outside the embedding space

Theorem 4: Impossibility of evolution outside the embedding space carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.35 Theorem 5: Necessity

Theorem 5: Necessity and sufficiency of compatibility with MM carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.36 Theorem 6: Structural tension

Theorem 6: Structural tension and phase transitions carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.37 Theorem 7: Universal law of complexity growth

Theorem 7: Universal law of complexity growth carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.38 Theorem 8: Impossibility of eternal stabilisation

Theorem 8: Impossibility of eternal stabilisation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.39 Theorem 10: Monotonic growth of embedding spaces

Theorem 10: Monotonic growth of embedding spaces carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.40 Theorem 12: Incompleteness of embedding spaces

Theorem 12: Incompleteness of embedding spaces carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.41 Proof-bearing

Proof-bearing and nonclaim layers carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.42 Theorem Roadmap

Theorem Roadmap and Proof Navigation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.43 Load-bearing theorem spine

Load-bearing theorem spine carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.44 Theorem fate correction after external review

Theorem fate correction after external review carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.45 Proof Machinery, Revision Law, Competition,

Proof Machinery, Revision Law, Competition, and Compression carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.46 (F11.10) Failure of global coherence theorems

(F11.10) Failure of global coherence theorems carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.47 Proof obligations

Proof obligations carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.48 Promoted theorem-native claim

Promoted theorem-native claim carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.49 Proof Construction, Verification,

Proof Construction, Verification, and Proof-Flows carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.50 Inference, Deduction,

Inference, Deduction, and Proof Processes carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.51 Master Theorems of Core 1.2

Master Theorems of Core 1.2 carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.52 Representability Theorems for Levels (K_2)K_2, (K_4)K_4, (K_8)K_8

Representability Theorems for Levels (K_2)K_2, (K_4)K_4, (K_8)K_8 carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.53 Theorem 1 (Representability of (K₂)K₂)

Theorem 1 (Representability of (K₂)K₂) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.54 Theorem 2 (Representability of (K₄)K₄)

Theorem 2 (Representability of (K₄)K₄) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.55 Theorem 3 (Representability of (K₈)K₈)

Theorem 3 (Representability of (K₈)K₈) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.56 Compatibility Theorem (K M)K M

Compatibility Theorem (K M)K M carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.57 Theorem 4 (Necessary

Theorem 4 (Necessary and sufficient conditions for compatibility (K M)K M) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.58 Theorem on Impossibility of Evolution Outside (M)M

Theorem on Impossibility of Evolution Outside (M)M carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.59 Theorem 5 (Impossibility of evolution outside the meta-domain)

Theorem 5 (Impossibility of evolution outside the meta-domain) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.60 Theorem on Death of a Continuum

Theorem on Death of a Continuum carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.61 Theorem 7 (Equivalence of death conditions)

Theorem 7 (Equivalence of death conditions) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.62 Theorem on Phase Transition

Theorem on Phase Transition carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.63 Theorem on Impossibility of Degradational Evolution

Theorem on Impossibility of Degradational Evolution carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.64 Source Witnessing, Theorem Fate,

Source Witnessing, Theorem Fate, and Publication Boundary carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.65 Executable Finite-Model Evidence

Executable Finite-Model Evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.66 Prediction

Prediction and theorem evidence boundaries carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.67 Hybrid-Escalation Evidence Bar

Hybrid-Escalation Evidence Bar and Execution Protocols carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.68 Global evidence bar

Global evidence bar carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.69 (P10.3) Proof-Flow Emergence

(P10.3) Proof-Flow Emergence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.70 P4: Predictions for Proof/Computation Flows J^{10}

P4: Predictions for Proof/Computation Flows J^{10} carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.71 (P10.12) Proof Flows Have Attractor Structure

(P10.12) Proof Flows Have Attractor Structure carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.72 P1: Predictions About the Birth of Model-Spaces

P1: Predictions About the Birth of Model-Spaces carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.73 (P11.2) Model-Space Dimensional Trigger

(P11.2) Model-Space Dimensional Trigger carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.74 (P11.5) Universal Axes of Model-Spaces

(P11.5) Universal Axes of Model-Spaces carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.75 (P11.6) Existence of Model-Topologies

(P11.6) Existence of Model-Topologies carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.76 (P11.17) The Model-Theoretic Cycle

(P11.17) The Model-Theoretic Cycle carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.77 (P11.21) Model-Space Fragmentation Reduces $k(K_{11})$

(P11.21) Model-Space Fragmentation Reduces $k(K_{11})$ carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.78 (P12.1) Dynamization Threshold of Model-Spaces

(P12.1) Dynamization Threshold of Model-Spaces carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.79 (P12.24) Fragmentation of Model-Dynamics Reduces $k(K_{12})$

(P12.24) Fragmentation of Model-Dynamics Reduces $k(K_{12})$ carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.80 (P12.26) No Stable Termination of Model-Dynamics

(P12.26) No Stable Termination of Model-Dynamics carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.81 (P9.12) Proof-like Flows Are Emergent

(P9.12) Proof-like Flows Are Emergent carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.82 (P9.26) Birth of Proof-Flow Attractors

(P9.26) Birth of Proof-Flow Attractors carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.83 Target-Blind Numeric Evidence

Target-Blind Numeric Evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.84 OC 1.3.3 Comparator

OC 1.3.3 Comparator and Novelty Matrix is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for OC 1.3.3 Comparator and Novelty Matrix. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.85 Prior-Art Interpretation for Article Readers

Prior-Art Interpretation for Article Readers is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Prior-Art Interpretation for Article Readers. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.86 Prior-Art Comparator

Prior-Art Comparator and Novelty Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Prior-Art Comparator and Novelty Boundary. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.87 Prior-Art

Prior-Art and Literature Position is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Prior-Art and Literature Position. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.88 Master Monograph Appendix: Comparator And Novelty Matrix

Master Monograph Appendix: Comparator And Novelty Matrix is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Master Monograph Appendix: Comparator And Novelty Matrix. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.89 OC133-NOVELTY-001

OC133-NOVELTY-001 is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for OC133-NOVELTY-001. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.90 appendix/OC_1_3_3_COMPARATOR_AND_NOVELTY_MATRIX.tex

Appendix/OC_1_3_3_COMPARATOR_AND_NOVELTY_MATRIX.tex is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for appendix/OC_1_3_3_COMPARATOR_AND_NOVELTY_MATRIX.tex. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.91 Reviewer challenge 7: Prior Art Positioning

Reviewer challenge 7: Prior Art Positioning states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.92 Prior-Art Historian

Prior-Art Historian is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Prior-Art Historian. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.93 Worked Attack Transcript C: Prior-Art Surface

Worked Attack Transcript C: Prior-Art Surface states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.94 Reviewer challenge 6: Phenomenon Coverage Model Card

Reviewer challenge 6: Phenomenon Coverage Model Card states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.95 Practical Utility

Practical Utility and Model Comparison Atlas is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Practical Utility and Model Comparison Atlas. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.96 Serious model comparison matrix

Serious model comparison matrix is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Serious model comparison matrix. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.97 Serious model comparison trace matrix

Serious model comparison trace matrix is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Serious model comparison trace matrix. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.98 10. T133-KLEVEL: Declared adjacent K-level atlas/evaluator consistency theorem

10. T133-KLEVEL: Declared adjacent K-level atlas/evaluator consistency theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.99 Reviewer challenge 1: Theorem Proof Binding

Reviewer challenge 1: Theorem Proof Binding carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.100 Reviewer challenge 2: Theorem Public Surface Binding

Reviewer challenge 2: Theorem Public Surface Binding carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.101 Attack Family 5: Public Wording Outruns Evidence

Attack Family 5: Public Wording Outruns Evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.102 Evidence-to-Wording Calibration

Evidence-to-Wording Calibration carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does

not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.103 Claim-Family Evidence Recipes

Claim-Family Evidence Recipes carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.104 Worked Attack Transcript A: Theorem Surface

Worked Attack Transcript A: Theorem Surface carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.105 Proof Machinery Appendix

Proof Machinery Appendix carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.106 Alternative-model competition matrix

Alternative-model competition matrix synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.107 Load-bearing proof tasks

Load-bearing proof tasks carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote

stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.108 Closure priority board

Closure priority board synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.109 Ontological Structure of the Model

Ontological Structure of the Model synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.110 Theorem (global verdict-invariant minimality)

Theorem (global verdict-invariant minimality) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.111 Proof

Proof carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.112 Theorem 1: Monotonicity of dimensionality

Theorem 1: Monotonicity of dimensionality carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.113 Theorem 2: Impossibility of spontaneous dimension creation

Theorem 2: Impossibility of spontaneous dimension creation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.114 Theorem 3: Death through boundary

Theorem 3: Death through boundary and embedding collapse carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.115 Theorem 4: Impossibility of evolution outside the embedding space

Theorem 4: Impossibility of evolution outside the embedding space carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.116 Theorem 5: Necessity

Theorem 5: Necessity and sufficiency of compatibility with MM carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.117 Theorem 6: Structural tension

Theorem 6: Structural tension and phase transitions carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.118 Theorem 7: Universal law of complexity growth

Theorem 7: Universal law of complexity growth carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.119 Theorem 8: Impossibility of eternal stabilisation

Theorem 8: Impossibility of eternal stabilisation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.120 Theorem 9: Death as loss of cycles

Theorem 9: Death as loss of cycles carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.121 Theorem 10: Monotonic growth of embedding spaces

Theorem 10: Monotonic growth of embedding spaces carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.122 Theorem 11: Threshold-expressivity incompatibility

Theorem 11: Threshold-expressivity incompatibility carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.123 Theorem 12: Incompleteness of embedding spaces

Theorem 12: Incompleteness of embedding spaces carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.124 Patch Model of Boundaries

Patch Model of Boundaries synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.125 Proof-bearing

Proof-bearing and nonclaim layers carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.126 Theorem Roadmap

Theorem Roadmap and Proof Navigation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.127 Load-bearing theorem spine

Load-bearing theorem spine carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.128 Theorem fate correction after external review

Theorem fate correction after external review carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.129 Worked Example 1: collapse, residue,

Worked Example 1: collapse, residue, and rebirth in a minimal admissible-state model synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.130 Proof Machinery, Revision Law, Competition,

Proof Machinery, Revision Law, Competition, and Compression carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.131 Alternative-model competition

Alternative-model competition synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.132 Internal models

Internal models synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.133 Type~IV Experiments: Internal Models

Type~IV Experiments: Internal Models and Coherence Testing synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.134 Type~III Experiments: Meta-Model Dynamics

Type~III Experiments: Meta-Model Dynamics and Explanatory Power synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.135 (F10.27) Meta-model cycle collapse

(F10.27) Meta-model cycle collapse synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.136 (F11.10) Failure of global coherence theorems

(F11.10) Failure of global coherence theorems carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.137 (F6.14) Negative model confidence

(F6.14) Negative model confidence synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.138 (F6.33) Broken model cycle

(F6.33) Broken model cycle synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on

historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.139 (F6.42) Model-collapse at birth

(F6.42) Model-collapse at birth synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.140 (F9.15) Inability to integrate cross-domain evidence

(F9.15) Inability to integrate cross-domain evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.141 (F9.28) Breakdown of the model cycle

(F9.28) Breakdown of the model cycle synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.142 (F9.36) No emergence of formal models

(F9.36) No emergence of formal models synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.143 Serious model families

Serious model families synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.144 What prior science could do, what remained fragmented,

What prior science could do, what remained fragmented, and what OC closes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.145 Proof obligations

Proof obligations carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.146 Promoted theorem-native claim

Promoted theorem-native claim carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.147 Jets of Meta-Modelling Operators F

Jets of Meta-Modelling Operators F synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.148 Jets of Categories of Models C

Jets of Categories of Models C synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.149 Patch Model Jets

Patch Model Jets and Local Instabilities synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.150 Jets of Model Spaces M

Jets of Model Spaces M synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.151 Role of kk-Level Structure in the Core Model

Role of kk-Level Structure in the Core Model synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.152 Proof Construction, Verification,

Proof Construction, Verification, and Proof-Flows carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.153 Proto-Representation

Proto-Representation and Internal Modelling Processes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.154 Model Building

Model Building and Theorisation Processes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis

draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.155 Inference, Deduction,

Inference, Deduction, and Proof Processes carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.156 Master Theorems of Core 1.2

Master Theorems of Core 1.2 carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.157 Representability Theorems for Levels (K_2)K_2, (K_4)K_4, (K_8)K_8

Representability Theorems for Levels (K_2)K_2, (K_4)K_4, (K_8)K_8 carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.158 Theorem 1 (Representability of (K_2)K_2)

Theorem 1 (Representability of (K_2)K_2) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.159 Theorem 2 (Representability of (K_4)K_4)

Theorem 2 (Representability of (K_4)K_4) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay

material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.160 Theorem 3 (Representability of (K_8)K_8)

Theorem 3 (Representability of (K_8)K_8) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.161 Compatibility Theorem (K M)K M

Compatibility Theorem (K M)K M carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.162 Theorem 4 (Necessary)

Theorem 4 (Necessary and sufficient conditions for compatibility (K M)K M) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.163 Theorem on Impossibility of Evolution Outside (M)M

Theorem on Impossibility of Evolution Outside (M)M carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.164 Theorem 5 (Impossibility of evolution outside the meta-domain)

Theorem 5 (Impossibility of evolution outside the meta-domain) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.165 Theorem on Birth of a New Dimension via ()

Theorem on Birth of a New Dimension via () carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.166 Theorem 6 (Birth of a new dimension)

Theorem 6 (Birth of a new dimension) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.167 Theorem on Death of a Continuum

Theorem on Death of a Continuum carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.168 Theorem 7 (Equivalence of death conditions)

Theorem 7 (Equivalence of death conditions) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.169 Theorem on Phase Transition

Theorem on Phase Transition carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.170 Theorem on Impossibility of Degradational Evolution

Theorem on Impossibility of Degradational Evolution carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.171 Theorem 9 (No “backward” dimensional reduction for a living continuum)

Theorem 9 (No “backward” dimensional reduction for a living continuum) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.172 Evidence Summary

Evidence Summary carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.173 Theorem

Theorem and Proof Registry carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.174 1. T133-K0-RES: K0 resolution-relative distinguishability theorem

1. T133-K0-RES: K0 resolution-relative distinguishability theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.175 2. T133-OMEGA-STATUS: Typed liveness, death, residue,

2. T133-OMEGA-STATUS: Typed liveness, death, residue, and rebirth evidence-consistency theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.176 3. T133-K-ZERO: Continuumness zero obstruction theorem

3. T133-K-ZERO: Continuumness zero obstruction theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.177 4. T133-BOUNDARY: Metric-threshold boundary specialization theorem

4. T133-BOUNDARY: Metric-threshold boundary specialization theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.178 5. T133-HYBRID: Typed update

5. T133-HYBRID: Typed update and chart-labelled operator semantics theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the

source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.179 6. T133-DIM: Historical axis

6. T133-DIM: Historical axis and effective-rank compatibility theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.180 7. T133-CYCLE: Live-status cycle-mode requirement theorem

7. T133-CYCLE: Live-status cycle-mode requirement theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.181 8. T133-ID: Identity, residue,

8. T133-ID: Identity, residue, and rebirth evidence-classification theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.182 9. T133-MIN: Declared semantic-verdict component independence theorem

9. T133-MIN: Declared semantic-verdict component independence theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.183 Does OC actually explain operators in non-smooth proof

Does OC actually explain operators in non-smooth proof and rewrite domains? carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.184 Does OC actually explain theory change as live claim/evidence update?

Does OC actually explain theory change as live claim/evidence update? carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.185 Theorem

Theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.186 Master Monograph Appendix: Boundary Representation Theorem

Master Monograph Appendix: Boundary Representation Theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.187 Boundary Representation Theorem

Boundary Representation Theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it

does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.188 Master Monograph Appendix: Verdict Invariant Minimality Full Proof

Master Monograph Appendix: Verdict Invariant Minimality Full Proof carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.189 OC 1.3.3 Verdict-Invariant Minimality Full Proof

OC 1.3.3 Verdict-Invariant Minimality Full Proof carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.190 FM-T133-HYBRID-PROOF-UPDATE-POS

FM-T133-HYBRID-PROOF-UPDATE-POS carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.191 FM-T133-HYBRID-PROOF-UPDATE-NEG

FM-T133-HYBRID-PROOF-UPDATE-NEG carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.192 FM-T133-HYBRID-PROOF-NO-RULE-NEG

FM-T133-HYBRID-PROOF-NO-RULE-NEG carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The

support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.193 appendix/OC_1_3_3_BOUNDARY_REPRESENTATION_THEOREM.tex

Appendix/OC_1_3_3_BOUNDARY_REPRESENTATION_THEOREM.tex carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.194 appendix/OC_1_3_3_VERDICT_INVARIANT_MINIMALITY_FULL_PROOF.tex

Appendix/OC_1_3_3_VERDICT_INVARIANT_MINIMALITY_FULL_PROOF.tex carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.195 Evidence Inputs

Evidence Inputs and Outputs carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.196 Evidence Reading Examples

Evidence Reading Examples carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.197 Proof Machinery Appendix

Proof Machinery Appendix carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator

evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.198 Alternative-model competition matrix

Alternative-model competition matrix is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Alternative-model competition matrix. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.199 Boundary Representation Theorem

Boundary Representation Theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.200 Theorem

Theorem carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.201 Proof

Proof carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.202 OC 1.3.3 Verdict-Invariant Minimality Full Proof

OC 1.3.3 Verdict-Invariant Minimality Full Proof carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay

material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.203 Load-bearing proof obligations

Load-bearing proof obligations carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.204 Closure priority board

Closure priority board is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Closure priority board. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.205 Definitions Before Evidence IDs

Definitions Before Evidence IDs carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.206 Evidence

Evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.207 Results

Results and Evidence Interpretation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material,

comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.12.208 (F9.15) Inability to integrate cross-domain evidence

(F9.15) Inability to integrate cross-domain evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

2.12.209 What prior science could do, what remained fragmented,

What prior science could do, what remained fragmented, and what OC closes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for What prior science could do, what remained fragmented, and what OC closes. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.13 Source Trace

Exact source bindings, path hashes, quality scorer hooks, and transition records are recorded in the review package manifest. They are kept out of the main prose to avoid turning the document into a ledger dump.